PAPER_679: Aether Superfluid Dynamics: UQFF Universal Aether as a Bosonic Condensate

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Abstract

The UQFF Universal Aether [UA] is modelled as a coherent bosonic superfluid described by a macroscopic wavefunction $\Psi(r,t) = \sqrt{n_{\mathrm{UA}}}e^{i\phi}$. We derive the healing length $\xi_{\mathrm{UA}} = \hbar/\sqrt{2m_{\mathrm{UA}}g_{\mathrm{UA}}n_{\mathrm{UA}}}$, sound speed $c_{\mathrm{UA}} = \sqrt{g_{\mathrm{UA}}n_{\mathrm{UA}}/m_{\mathrm{UA}}}$, and effective gravitational coupling $g_{\mathrm{eff}}(r) = g_N(r)(1 + c_{\mathrm{UA}}^2/c^2 \cdot f_{\mathrm{TRZ}}\rho_{\mathrm{UA}}/\rho_{\mathrm{SCm}})$.

Primary UQFF Equation

$$c_{\mathrm{UA}} = \sqrt{\frac{g_{\mathrm{UA}}n_{\mathrm{UA}}}{m_{\mathrm{UA}}}}, \quad \xi_{\mathrm{UA}} = \frac{\hbar}{\sqrt{2m_{\mathrm{UA}}g_{\mathrm{UA}}n_{\mathrm{UA}}}}$$

Parameters

Parameter	Value	Description
m_{UA}	$\sim 2 \times 10^{-68}$ kg	Ultralight boson mass
ρ_{UA}	7.09×10^{-36} kg/m^3	Aether density
g_{UA}	10^{-10} m^3/s^2	Interaction strength

UQFF Constants

Constant	Value	Description
ρ_{UA}	7.09×10^{-36} kg/m^3	Universal Aether density
ρ_{SCm}	7.09×10^{-37} kg/m^3	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	5×10^{-4} day^{-1}	UQFF calibration constant
$[\mathrm{SSq}]$	0.57	Superstring quenching factor
μ_J	3.38×10^{23} J·m	Magnetic string coupling
γ	5×10^{-5} / 86400 s^{-1}	Aether oscillation decay

C++ Implementation

```cpp
#include "AetherSuperfluidDynamics.h"

AetherSuperfluidDynamics obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```python
from CondensedPhysics2 import AetherSuperfluidDynamicsCalculator

calc = AetherSuperfluidDynamicsCalculator()
result = calc.compute()  # returns all UQFF quantities
print(result)
```

References

Penrose 1996 (quantum collapse); Sudarsky 2020 (BEC dark matter); UQFF Session 173

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